

CompTIA A+ Assignment

❖ Module -3, 4 [Installation and Maintenance of Hardware and Its components]

1. What is User Management?

- ❖ User management is the process of creating, modifying, deleting, and controlling user accounts and permissions in an operating system.

2. Where can we access User Management?

❖ In Windows:

- Press Windows + R
- Type: lusrmgr.msc
- Press Enter

Or:

- Right Click This PC → Manage → Local Users and Groups

3. Why is User Management needed?

- To provide security
- To control user access
- To manage permissions
- To protect files and resources
- To create separate accounts for different users

4. Practical: Create a User from User Management

❖ Steps:

- Open lusrmgr.msc
- Click Users
- Right click → New User
- Enter:
- Click Create
- Click Close

5. Practical: Change Administrator Password

❖ Steps:

- Open lusrmgr.msc
- Click Users
- Right click Administrator
- Select Set Password
- Enter new password
- Click OK

6. What is File and Folder Permission?

❖ Permissions control access to files and folders.

Types:

- Sharing Permission: Applied when folder is shared over network
- NTFS Permission: Applied locally and over network, more secure and detailed

7. Use of File and Folder Permission

- Protect data
- Restrict unauthorized access
- Allow read/write/modify access
- Improve security

8. Steps to Give Folder Read Only Permission

- Right click folder
- Select Properties
- Open Security tab
- Click Edit
- Select user
- Check Read
- Uncheck Write/Modify
- Click Apply → OK

9. Steps to Give File Only Admin Permission

- Right click file → Properties
- Open Security
- Click Advanced
- Disable inheritance
- Remove other users
- Add Administrator
- Give Full Control
- Click Apply

10. Practical: Change Ownership of Folder and Subfolders

- Right click folder → Properties
- Security → Advanced
- Click Change beside Owner
- Enter new username
- Check:
- Click Apply → OK

11. What is OS?

❖ Operating System (OS) is system software that manages hardware and software resources.

Examples:

- Windows
- Linux
- macOS

12. Types of OS

- Single User OS
- Multi User OS
- Network OS
- Real Time OS
- Distributed OS
- Mobile OS

13. Practical: Create Bootable Kali Linux USB & Install

- ❖ Using Rufus:
 - Download Kali ISO
 - Insert USB drive
 - Open Rufus
 - Select USB
 - Select Kali ISO
 - Click Start
 - Boot from USB
 - Follow installation steps

14. Practical: Create Windows 10 Bootable USB & Install

- Download Windows 10 ISO from Microsoft Windows 10 Download
- Open Rufus
- Select ISO and USB
- Click Start
- Boot system using USB
- Install Windows

15. Create macOS Big Sur Bootable USB using UniBeast

- ❖ Download:
 - macOS Big Sur
 - UniBeast
- Format USB as Mac OS Extended
- Open UniBeast
 - Select macOS Big Sur
 - Create bootable USB

16. What is Clean Install?

- ❖ Installing OS by removing previous OS and formatting drive.

17. What is Upgrade Installation?

- ❖ Installing a newer version of OS while keeping files and applications.

18. Practical: Upgrade Windows 8 to Windows 10

- Backup data
- Download Windows 10 setup
- Run setup.exe
- Choose:
- Follow installation instructions

19. What is Partitioning?

- ❖ Dividing a hard disk into separate sections.

20. What is Partition?

- ❖ A partition is a logical division of storage space.

21. What is Format?

- ❖ Formatting prepares storage media for use by creating a file system.

22. Format Partition Using CMD

- diskpart
- list disk
- select disk 0
- list partition
- select partition 1
- format fs=ntfs quick

23. Administrative Tools

- Disk Management
- Device Manager
- Event Viewer
- Services
- Task Scheduler
- Computer Management
- Local Security Policy
- Performance Monitor

24. What is Disk Management Tool?

- ❖ A Windows utility used to manage disks and partitions.

25. Operations in Disk Management

- Create partition
- Delete partition
- Format partition
- Extend/Shrink volume
- Change drive letter

26. Practical: Delete Driver from Device Manager

- Open Device Manager
- Expand device category
- Right click device
- Click Uninstall Device
- Confirm removal

27. What is Windows Features?

- ❖ Optional Windows components that can be enabled or disabled.

28. Install .NET Framework 3.5

- ❖ Open:
 - Control Panel → Programs
- Click:
 - Turn Windows features on or off
- Check:
 - .NET Framework 3.5
- Click OK

29. Disable Internet Explorer

❖ Open Windows Features

- Uncheck:
- Click OK
- Restart PC

30. What is Backup and Restore?

- Backup: Copying data for safety
- Restore: Recovering backed-up data

31. Tools of Backup

- Windows Backup
- System Restore
- File History
- Third-party backup software

32. Restore from Restore Point

- Search: rstrui.exe
- Select restore point
- Click Next
- Restore system

33. Protection from Electrical Fluctuation

❖ Use:

- UPS
- Stabilizer
- Surge protector

34. What is OS Based Firewall?

❖ A firewall built into operating system to control incoming and outgoing traffic.

Example:

Windows Defender Firewall

35. Configure Inbound and Outbound Rules

- ❖ Open:
 - wf.msc
 - Select Inbound/Outbound Rules
 - Click New Rule
 - Configure ports/programs

36. Practical: Block Internet with Firewall

- ❖ Open Windows Firewall
 - Outbound Rules → New Rule
 - Select:
 - Choose:
 - Apply to all profiles
 - Finish

Module 4

37. What is Troubleshooting?

- ❖ A process used to identify and fix computer problems.

38. Change Forgotten Administrator Password

- ❖ Using CMD:

```
net user administrator newpassword
```

39. Troubleshoot No Display Problem

- Check monitor power
- Check VGA/HDMI cable
- Reseat RAM
- Check GPU
- Test another monitor

40. Blue Screen of Death (BSOD)

❖ Causes:

- Driver failure
- RAM issues
- Corrupted OS

Solutions:

- Safe Mode
- Update drivers
- Run memory diagnostics
- System Restore

41. Practical: Repair OS

❖ sfc /scannow

DISM /Online /Cleanup-Image /RestoreHealth

42. Practical: Repair Boot File

- bootrec /fixmbr
- bootrec /fixboot
- bootrec /rebuildbcd

43. Recover Deleted File

- Use Recycle Bin
- Use recovery software like : Recuva

44. Recover Formatted File

❖ Use:

- Recuva
- EaseUS
- Disk Drill

45. Recover Data from Corrupted OS

- Boot using recovery USB
- Copy data using Linux Live USB
- Use recovery software

46. Basic Printer Troubleshooting

- Check power
- Check cables
- Check paper jam
- Reinstall drivers
- Check ink/toner

47. Basic Laptop Troubleshooting

- Check charger
- Remove battery
- Test RAM
- Clean overheating dust
- Check display

48. Laptop Not Starting Practical

- Check adapter
- Remove battery
- Hold power button 30 sec
- Reseat RAM
- Test motherboard

49. Disassemble Laptop & Change Corrupted

- Power off laptop
- Remove battery
- Open back panel
- Remove faulty RAM
- Insert new RAM
- Reassemble laptop

50. Change Printer Cartridge

- Open printer cover
- Remove old cartridge
- Unpack new cartridge
- Install properly
- Close cover